

CHILL ZEN in brief: generative AI from noisy memory reads

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This is a condensed account of a longer technical paper on CHILL ZEN (Comparator-Heated In-memory Lane Learning with Zero-Ended Noise), an architecture that generates images inside memristor memory instead of on a processor. Each image is produced by about 3600 noisy reads of stored weights, with no equilibrium sampling phase. The randomness comes mostly from the read comparator, whose noise is set by each read instruction, so the same memory reads “hot” to draw and “cold” to decide. In emulation on binarized Fashion-MNIST the system scores FID 18.1, or 9.7 when trained on binary images, against 24.9 for the published eight-step Denoising Thermodynamic Model (DTM). Under the stated energy models it uses about one eighth of the DTM’s energy per image. All results are emulated; the lane-scale hardware has not been built. We state what has been shown, what is modeled, and what remains to be demonstrated.

I. WHY THIS MATTERS

Generative AI is expensive because it samples. A diffusion model or an energy-based model produces each output through a long chain of randomized updates, and every update runs on a digital processor that must manufacture its own randomness from deterministic logic. The energy cost of that sampling has become a first-order constraint on deployment [1].

One response is to let physics do the sampling. Probabilistic bits built from stochastic nanomagnets and sub-threshold transistors have solved optimization and learning problems at very low energy [2–4]. The most complete proposal to date is the Denoising Thermodynamic Model and its hardware, the Denoising Thermodynamic Computer Architecture (DTCA) [5]. It chains two to eight energy-based models, each sampled by a bank of transistor random-number generators, and it reports both image quality and an energy budget. That paper also names the central difficulty. A single energy-based model that fits real data well develops deep energy barriers between its modes, and any local sampler then takes a long time to mix between them. Chaining models keeps each step easy to mix, but the chain still needs on the order of ten million random updates per image.

CHILL ZEN takes a different route to the same goal. Instead of reversing a noising process through equilibrium sampling, it draws an image’s discrete code in one fixed sequence of noisy memory reads and then reconciles the code in one cold sweep. There is no mixing phase. The randomness that makes each draw different is supplied at the point where the memory is read, by a comparator whose noise level is part of the read instruction. The full paper reports that this design, in emulation, beats the published DTM on quality and on modeled energy at once. This document explains how, and how much of that claim rests on measurement versus modeling.

II. THE IDEA IN ONE PAGE

A. The memory element is also the sampler

The storage element is a pair of memristors read through a voltage divider. The pair’s balance is the stored weight $w \in [-1, 1]$, and the pair’s total conductance is its magnitude m . Magnitude acts like accumulated evidence: the more a pair has been taught, the less a fixed feedback pulse moves it, and the less noise its read carries.

A read returns the weight plus noise. The noise has three sources. Two are in the devices, thermal and flicker, and both fall as magnitude grows. The third is the read comparator’s own input-referred noise, which does not depend on the stored state at all. The thermal term scales inversely with read voltage, so a low-voltage read is hot and a full-voltage read is cold. The sign of one hot read is a probabilistic bit whose bias is the stored weight. The same physical element stores the parameter and draws the sample.

B. Lanes pool pairs, and pooling quenches device noise

A neural lane pools many pairs on one shared divider. By Kirchhoff’s laws a lane reports the magnitude-weighted average of the selected weights. Selection is an address: an ordered tuple of symbols, one per address space, and each space contributes in proportion to its share of the pooled magnitude. Lanes are grouped into winner-take-all groups, one lane per symbol, and taught by a local three-case rule: read every lane, reinforce the correct one, oppose a wrong one that fired, and mildly reinforce a correctly silent one. There are no gradients and no host in the loop for the banks. Taught this way, a cold argmax readout is a classifier at parity with logistic regression on the same encoding.

Pooling has a consequence that determines the whole design. As hundreds of pairs are pooled, their device noise averages down to a floor set by the output node’s

capacitance, the familiar kT/C limit. At the assumed 10 mV hot read the lanes supply less than 40% of the noise a useful draw needs on the first read and less than 10% thereafter. The devices cannot heat themselves enough to sample. The comparator has to supply the temperature.

C. The comparator sets the temperature

Every comparator has input-referred noise, and circuit settings control it: lower the preamplifier bias, add an input noise source, or trigger the latch earlier. CHILL ZEN treats that setting as a trim register written by each read instruction, alongside read voltage and pulse width. A hot read uses low bias and a cold read uses full bias. The two regimes occur within a single image. At the operating point the comparator supplies more than 90% of the draw’s variance on every read after the first. This is what the name means: the lanes are heated by the comparator, and the noise is ended, zeroed, for the final decision.

D. The code and the generation procedure

Images are represented by an additive discrete code: a fixed set of codebooks whose selected atoms sum to the image through a fixed linear decoder. The code is fitted on the host with a dropout objective, so that partial codes decode to reasonable images and the address spaces carry information about one another. The deployed system uses a 904-bit code: a global backbone of 128 books over the whole canvas and a residual patch code on a 7×7 grid of slots, plus the clamped class label. Stripped of the hardware, the model class sits beside masked-token generation over a discrete codebook [6, 7], with local teaching in place of a transformer.

Each level pairs a soft generator bank, which draws with hot reads, and a hard repair bank, which corrects with one cold parallel sweep. Generation runs in five steps. Clamp the label. Draw the 128 backbone books one at a time with hot reads, then apply one cold repair sweep. Draw all 49 patch slots in one parallel hot read. Apply one cold joint sweep that reconciles both levels together. Decode through the fixed linear map. That is 131 sequential steps and about 3600 hot reads per image. The joint sweep runs exactly once by design: repeated doses pull every image toward its class prototype and lower quality, because the sweep’s fixed points are the prototypes and not the data.

III. WHAT WAS SHOWN

Every result is emulated lane computation on the open **ktram-neural-core** runtime [8]: device models with the read-noise physics above, every read and adapt issued as a lane instruction, and decoding and scoring in host

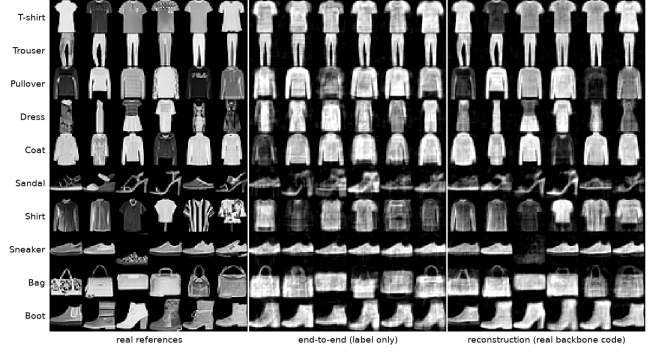


FIG. 1. **Emulated samples.** Real Fashion-MNIST references (left), end-to-end draws from the label alone (center), and the reconstruction control in which a real backbone code is supplied and only the patch level and joint sweep run (right). Six per class, the first six samples of one seed, with no selection.

floating point. The memristors and unit crossbars exist as fabricated, commercially sold parts [9]. The lane-scale system does not.

A. Grayscale generation

On grayscale Fashion-MNIST [10] the two-level system was scored by a frozen pixel classifier for label agreement, plus within-class diversity, tone spread, and contrast. End-to-end draws from the label alone score 0.90 on the critic against 0.89 for real images, with slightly higher diversity than real, lower tone spread, and contrast about 14% above real. The critic score is label agreement and not a claim that the images equal real ones. Over 5120 drawn images every code is distinct, none is a copy of a training image, and nearest-neighbor distances to the training set sit at or beyond those of held-out real images. The generator is not memorizing.

B. The comparator sweep

Sweeping the comparator’s noise setting with everything else fixed is the paper’s central experiment (Fig. 2). Every binary-trained configuration with comparator noise below 20% of the read voltage, including the device-only point, fails the diversity screen. Adding comparator noise improves both stacks, with the grayscale critic best at 20% of the read voltage and binary FID best at 20 to 30%. Raising the read voltage fivefold at fixed comparator ratio, which cuts lane thermal noise fivefold, leaves the metrics essentially unchanged. The comparator ratio, not the device physics, is the sampling temperature.

This yields a concrete design requirement for a circuit that has not yet been designed. The comparator’s input-referred noise must be adjustable from near zero to about

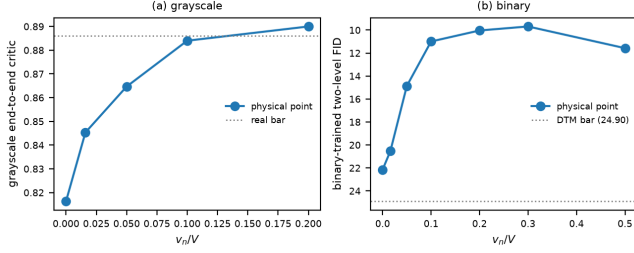


FIG. 2. **The comparator’s noise setting is the draw temperature.** (a) Grayscale critic score against comparator noise relative to read voltage, with the real-image score as a line. (b) The same sweep on the binary-trained stack scored as FID, with the published DTM value as a line. Lower FID is better.

TABLE I. Binarized Fashion-MNIST under the replication protocol. DTM rows are published values. Energies are modeled, not measured. Lower FID is better.

system	FID	J/image (modeled)
real fitting images (control)	1.9	—
CHILL ZEN two-level, grayscale-trained	18.1	2.0×10^{-9}
CHILL ZEN one-level, 128 books	22.1	7.5×10^{-10}
CHILL ZEN one-level, 64 books	22.6	2.5×10^{-10}
CHILL ZEN one-level, 32 books	25.8	9.9×10^{-11}
CHILL ZEN two-level, binary-trained	9.7	2.0×10^{-9}
DTM, 8 steps [5]	24.9	1.57×10^{-8}
DTM, 2 steps [5]	32.6	3.92×10^{-9}

20 to 30% of the read voltage, and its static offset must be trimmed below about 10% of the read voltage at every bias setting: 1 mV at a 10 mV read. A raw dynamic latch has the right noise range but carries 5 to 20 mV of offset, and an auto-zeroed or chopped comparator has the right offset but too little noise. Neither class meets both requirements on its own, which is why a per-lane trim plus a settable noise source is specified. A control run also showed that replacing the idealized noiseless cold reads with a physical noise floor flips some decisions but leaves every aggregate metric unchanged.

C. Head-to-head with the DTM on binarized Fashion-MNIST

The DTM paper reports quality as FID on binarized Fashion-MNIST but does not state its sample count, feature extractor, or binarization threshold, and its released library contains no FID code. That library links to a replication repository [11] with evaluation code and reference statistics, and we adopted that protocol without modification. As a check, all 60 000 real training images binarized at the replication’s threshold score FID 0.00004 against its shipped statistics, which fixes every element of the scoring path at once. The DTM values in Table I are transcribed from the published paper and were not rerun.

TABLE II. Modeled energy per image against the published eight-step DTM chain at 1.57×10^{-8} J. These are scenario calculations from the stated geometry, not measurements or uncertainty bounds.

operating point	J/image	DTM ratio
nominal: 4×4 array, 0.8 V select	2.0×10^{-9}	$7.9\times$
selector column, no sneak paths	1.6×10^{-9}	$9.9\times$
low-swing select, 0.5 V	8.8×10^{-10}	$17.7\times$
7 nm-class periphery, 0.5 V	4.2×10^{-10}	$37\times$
pessimistic corner	8.1×10^{-9}	$1.9\times$

Two results stand out. The grayscale-trained system, with weights unchanged and its renders simply thresholded, scores 18.1 against the DTM’s 24.9. Training the same architecture directly on binary images, which removes an information asymmetry in the grayscale path, lowers FID to 9.7 at the same modeled energy. The smaller one-level systems trace an energy-quality frontier: 128 and 64 books clear the DTM bar at a fraction of the energy, and 32 and 16 books do not. Drawing with the class chosen at random rather than clamped raises FID by less than half a point. A trivial independent-pixel sampler under the same protocol scores 180, so the protocol does discriminate.

For context, the DTM paper’s own baselines report 26.3 for a GAN, 17.9 for a VAE, and about 12 to 16 for a small diffusion model, none of which we could verify. CHILL ZEN’s 9.7 is in that company. One further caution deserves emphasis: the binarization threshold alone moves FID by more than the gap between any two systems. Real training images binarized at the midpoint rather than at 0.1 score 29.8 against the same reference, worse than the DTM. A number scored under an unstated data convention is not comparable to anything.

IV. ENERGY AND SPEED

Both systems get their randomness from a circuit at every node, so what separates them is how many random draws an image needs. CHILL ZEN uses about 3600 hot lane reads. The eight-step DTM chain at its stated 250 mixing iterations uses about 10^7 conditional updates. The two systems do different work per step, so the paper builds a physical energy model from a stated geometry: one 4×4 unit crossbar per address space on a 50 nm pitch, addresses broadcast on select lines across each bank, and one comparator per lane. Every constant is a stated dimension or a textbook value, with no fitted parameter. The model counts address delivery, divider reads, and readout, applied to the deployed system’s exact operation count of 10 064 lane reads. It includes the decoder, counted as the 784 cold lane reads it is designed to be.

At the nominal point the estimate is 2×10^{-9} J per image, about one eighth of the DTM chain (Table II).

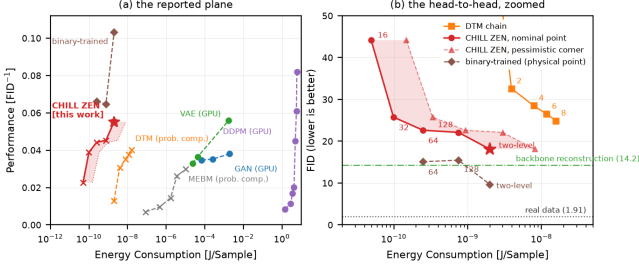


FIG. 3. **Quality against modeled energy.** (a) The axes of the DTM paper’s summary figure with its five published series re-plotted, and the CHILL ZEN frontier added: one-level systems of 16 to 128 books and the two-level system (star), with the shaded band extending each point to the model’s pessimistic corner. Binary-trained arms are diamonds. (b) The same frontier against the DTM chain alone, drawn as FID so the margin is readable. CHILL ZEN energies are model estimates.

Address delivery at logic voltage is 91% of that budget, which is the same communication term the DTM paper reports as about half of its own. Single-knob sensitivities move the ratio between about 4.6 and 7.9, and the pessimistic corner still leaves the DTM at about twice CHILL ZEN’s energy. The advantage comes from doing less sampling work, not from a cheaper random bit: the comparator’s roughly 10 fJ per read is more expensive than the DTCA’s measured 350 aJ per random bit. Fewer draws win by a wider margin than the per-draw cost loses.

On latency, the DTCA waits on its random-number generators, at about $400\mu\text{s}$ of sampling per image for the eight-step chain. CHILL ZEN needs 131 sequential steps plus one parallel decode. If each step fits the 62.5 ns window used to budget readout energy, an image takes about $8\mu\text{s}$, roughly fifty times faster. That window does not yet establish a complete instruction cycle, so it is an illustration and not a bound.

V. WHAT IS ESTABLISHED, MODELED, OR OPEN

The claims in this document rest on three different kinds of evidence, and a reader deciding what to make of them should keep the three apart.

Established in emulation. The architecture generates recognizable, diverse, non-memorized images from a fixed sequence of noisy reads with no equilibrium phase. Pooled device noise is insufficient for sampling and the comparator must supply the temperature. The comparator’s noise ratio behaves as a temperature knob with a clear optimum. Under a verified scoring path the emulated system beats the published DTM FID, at 18.1 or 9.7 against 24.9. The banks were taught by local lane instructions without gradients. The emulator is bit-exact against the reference lane core, and every number traces

to a script in the released repository.

Modeled, not measured. All energy figures come from a stated geometry with textbook constants. They exclude weight programming, die area, and leakage beyond a stated bound, as the DTM’s own accounting also excludes its programming. The nominal crossbar floats its unselected lines and therefore has sneak paths, which the emulator does not model; the selected symbol would contribute only 44% of the read conductance. The geometry the paper says it would build, a 16×1 selector column, has no sneak path and models at 1.6×10^{-9} J per image. The comparator is a specification and not a circuit. Its bandwidth, noise, and energy have not been demonstrated together in one design, and its kickback is unresolved. The lane-side decoder has not been emulated at memristor precision.

Open. The lane-scale system has not been fabricated. The emulator’s device noise constants are placeholders pending measurement. The reported scores are validation numbers: the evaluation partition informed epoch and noise-level selection, and each row is reported at its best comparator setting on a declared grid. The training splits differ from the DTM’s in size and membership. There is no gradient-trained baseline on the same code and connectivity, so the paper does not claim that the local rule outperforms gradient training. The remaining quality gap to real data, at FID 9.7 against a reconstruction reference of 2.8, is unexplained.

VI. BOTTOM LINE

CHILL ZEN reframes in-memory generative computing around a single observation: once weights are pooled on a lane, the memory is too quiet to sample itself, so the periphery must supply the temperature, and a comparator whose noise is set per instruction does that job at the point of readout. With that one knob, generation becomes a short, fixed sequence of hot reads and cold sweeps rather than an equilibrium chain, and the sampling count per image falls by more than three orders of magnitude relative to the published DTM. In emulation this delivers better FID than the DTM at about one eighth of its modeled energy, and the frontier of smaller systems shows the trade-off is tunable. What separates this from a product is a fabricated lane array with selector columns, a comparator that meets the stated noise and offset specification, and measured rather than modeled device constants. Those are the next steps, and each is a circuit-engineering task with a stated target rather than an open research question.

The full paper, code, saved evidence, and a map from every reported number to the script that produces it are prepared for release at <https://github.com/knownw/CHILL-ZEN>, with the lane emulator at <https://github.com/knownw/ktram-neural-core>. The kT-bit, the kT-RAM architecture, and its technology stack are the subject of granted U.S. patents [12–14], listed at <https://>

[//knowm.ai/ip/](https://knowm.ai/ip/). The author is part of Knowm Inc., which has made and sold memristors and plans to sell

products or intellectual property based on the technologies described here.

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- [1] T. Conte, E. DeBenedictis, N. Ganesh, T. Hylton, J. P. Strachan, R. S. Williams, A. Alemi, L. Altenberg, G. Crooks, J. Crutchfield, L. del Rio, J. Deutsch, M. DeWeese, K. Douglas, M. Esposito, M. Frank, R. Fry, P. Harsha, M. Hill, C. Kello, J. Krichmar, S. Kumar, S.-C. Liu, S. Lloyd, M. Marsili, I. Nemenman, M. A. Nugent, N. Packard, D. Randall, P. Sadowski, N. Santhanam, R. Shaw, A. Stieg, E. Stopnitzky, C. Teuscher, C. Watkins, D. Wolpert, J. Yang, and Y. Yufik, Thermodynamic computing, arXiv preprint arXiv:1911.01968 (2019).
 - [2] B. Sutton, K. Y. Camsari, B. Behin-Aein, and S. Datta, Intrinsic optimization using stochastic nanomagnets, *Scientific Reports* **7**, 44370 (2017).
 - [3] W. A. Borders, A. Z. Pervaiz, S. Fukami, K. Y. Camsari, H. Ohno, and S. Datta, Integer factorization using stochastic magnetic tunnel junctions, *Nature* **573**, 390 (2019).
 - [4] S. Niazi, S. Chowdhury, N. A. Aadit, M. Mohseni, Y. Qin, and K. Y. Camsari, Training deep Boltzmann networks with sparse Ising machines, *Nature Electronics* **7**, 610 (2024).
 - [5] A. Jelinčič, O. Lockwood, A. Garlapati, P. Schillinger, I. L. Chuang, G. Verdon, and T. McCourt, An efficient probabilistic hardware architecture for diffusion-like models, arXiv preprint arXiv:2510.23972v2 [10.48550/arXiv.2510.23972](https://arxiv.org/abs/2510.23972) (2025).
 - [6] H. Chang, H. Zhang, L. Jiang, C. Liu, and W. T. Freeman, MaskGIT: Masked generative image transformer, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2022).
 - [7] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, Neural discrete representation learning, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
 - [8] Knowm Inc., ktram-neural-core, <https://github.com/knowm/ktram-neural-core> (2026).
 - [9] Knowm Inc., Knowm self-directed channel memristors, https://knowm.org/downloads/Knowm_Memristors.pdf (2019), datasheet; Fig. 4 gives LRS/HRS versus write compliance current for W, Sn, and Cr devices.
 - [10] H. Xiao, K. Rasul, and R. Vollgraf, Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, arXiv preprint arXiv:1708.07747 (2017).
 - [11] **pschilli0range** and contributors, dtm-replication: a reference implementation and evaluation code for denoising thermodynamic models, <https://github.com/pschilli0range/dtm-replication> (2025), linked from the **thrml** README; the author credits a DTM author for guidance; commit **7c22d19**, accessed 2026-08.
 - [12] M. A. Nugent, Thermodynamic bit formed of two memristors (2015), U.S. Patent 9,099,179, granted 4 Aug. 2015; filed 5 Dec. 2013; assignee KnowmTech LLC, since merged into Knowm Inc.
 - [13] M. A. Nugent and T. W. Molter, Thermodynamic random access memory for neuromorphic computing utilizing AHaH (anti-Hebbian and Hebbian) and memristor components (2017), U.S. Patent 9,679,241, granted 13 Jun. 2017; filed 15 Aug. 2014; assignee KnowmTech LLC, since merged into Knowm Inc.
 - [14] M. A. Nugent and T. W. Molter, Thermodynamic-RAM technology stack (2019), U.S. Patent 10,311,357, granted 4 Jun. 2019; filed 27 May 2015; assignee KnowmTech LLC, since merged into Knowm Inc.